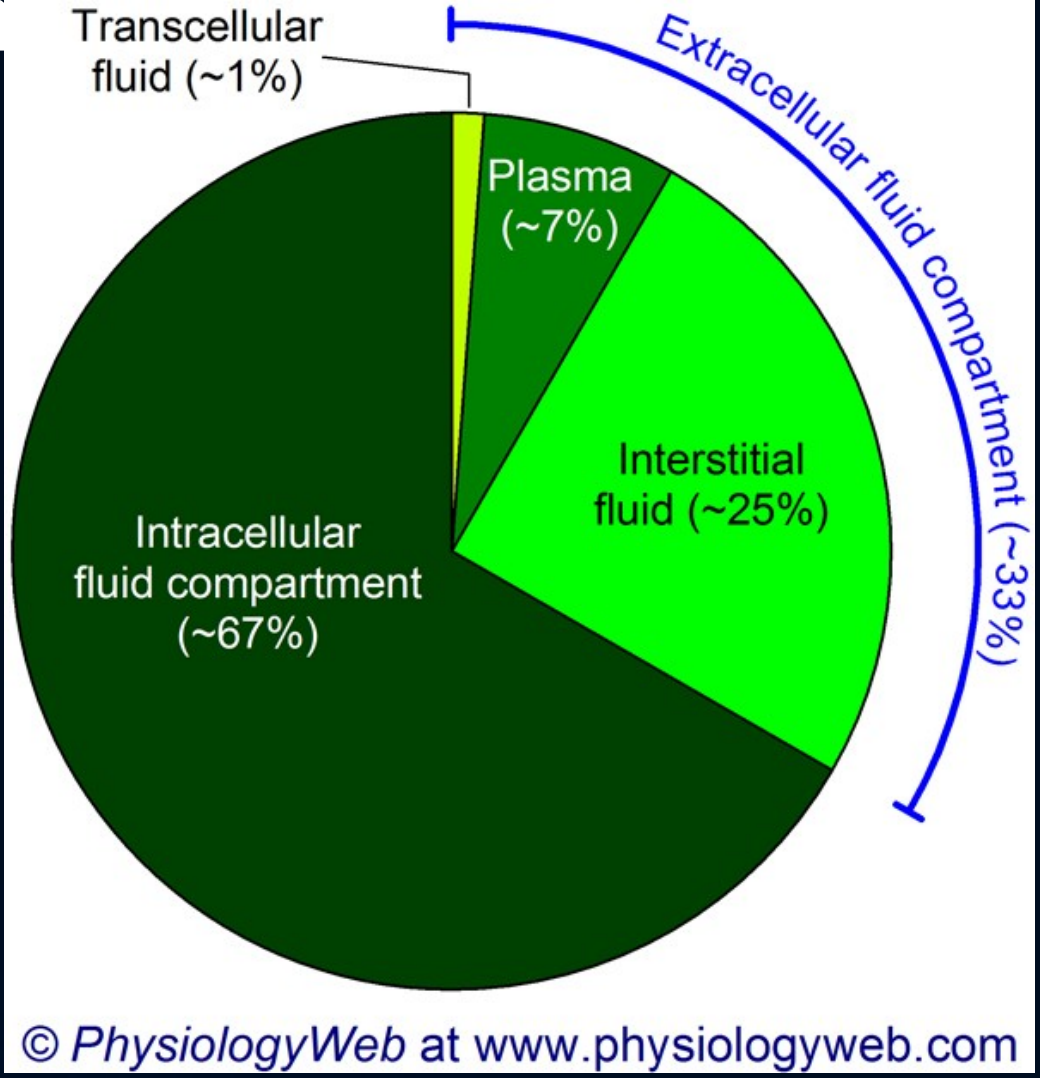
Renal Physiology

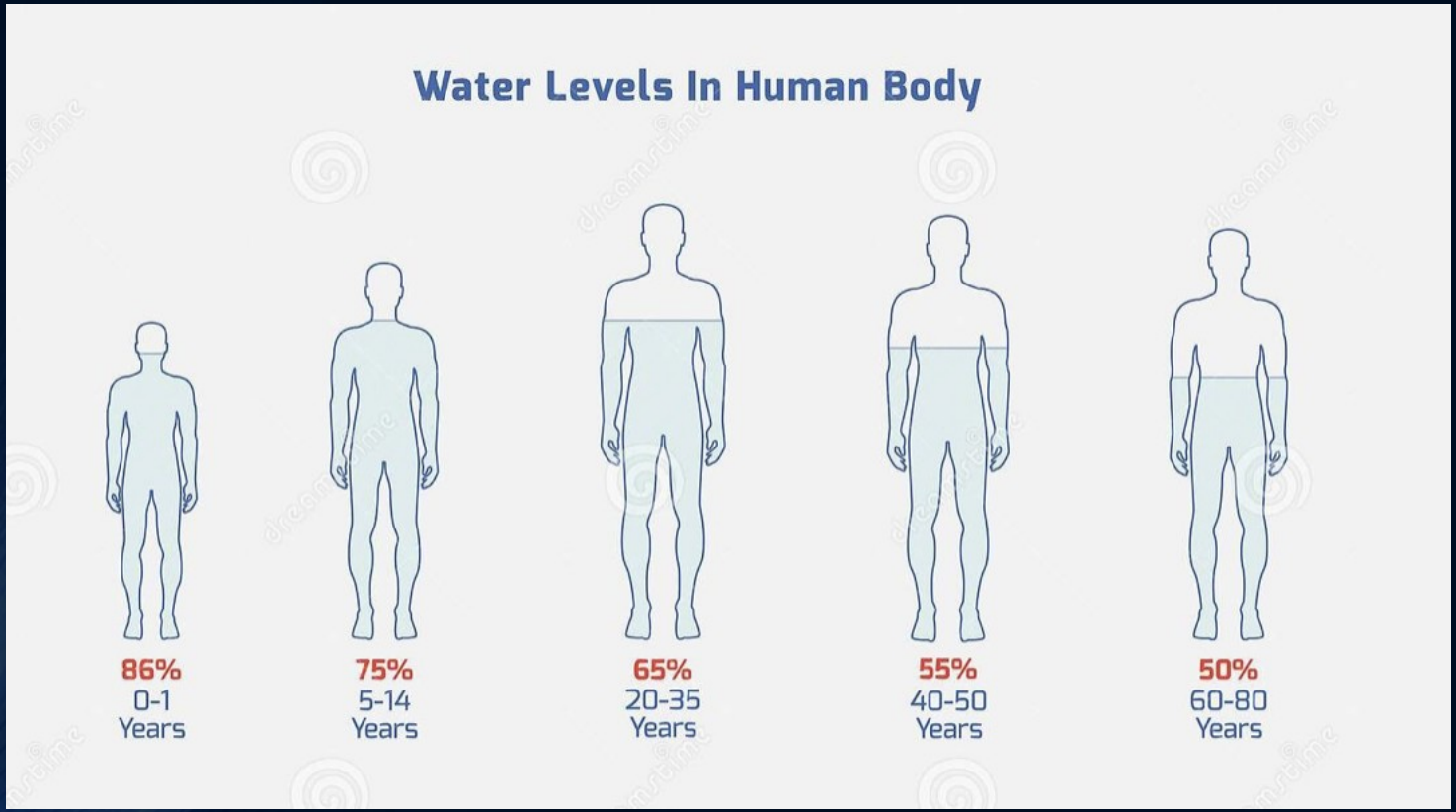
REGULATION OF FLUID AND BLOOD PRESSURE

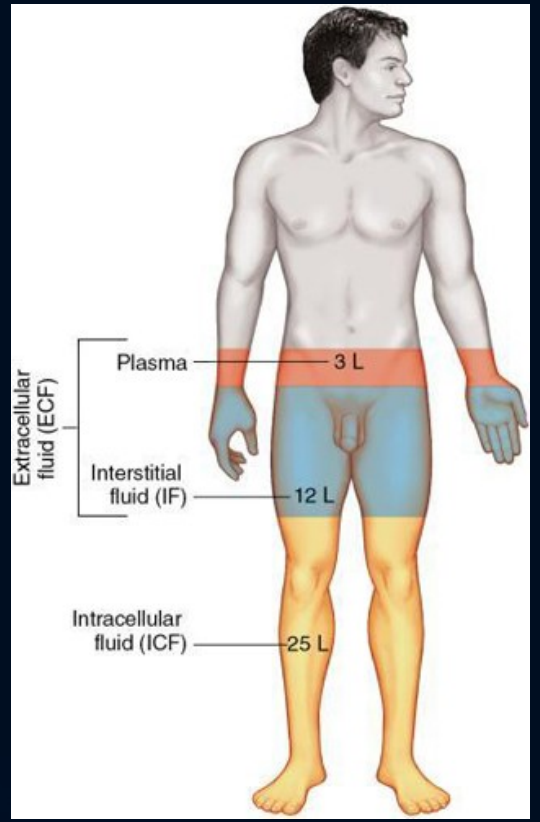
Regulation of Fluid Volum

- Fluid Compartments – volumes are based on a 70 kg male
- Intracellular fluid
  - Fluid inside cells – approximately 28 L
- Extracellular fluid
  - Interstitial fluid – 11 L
  - Plasma – 3 L
  - Transcellular fluid – less than 1 L



Fluid Distribution by Age & by Compartment





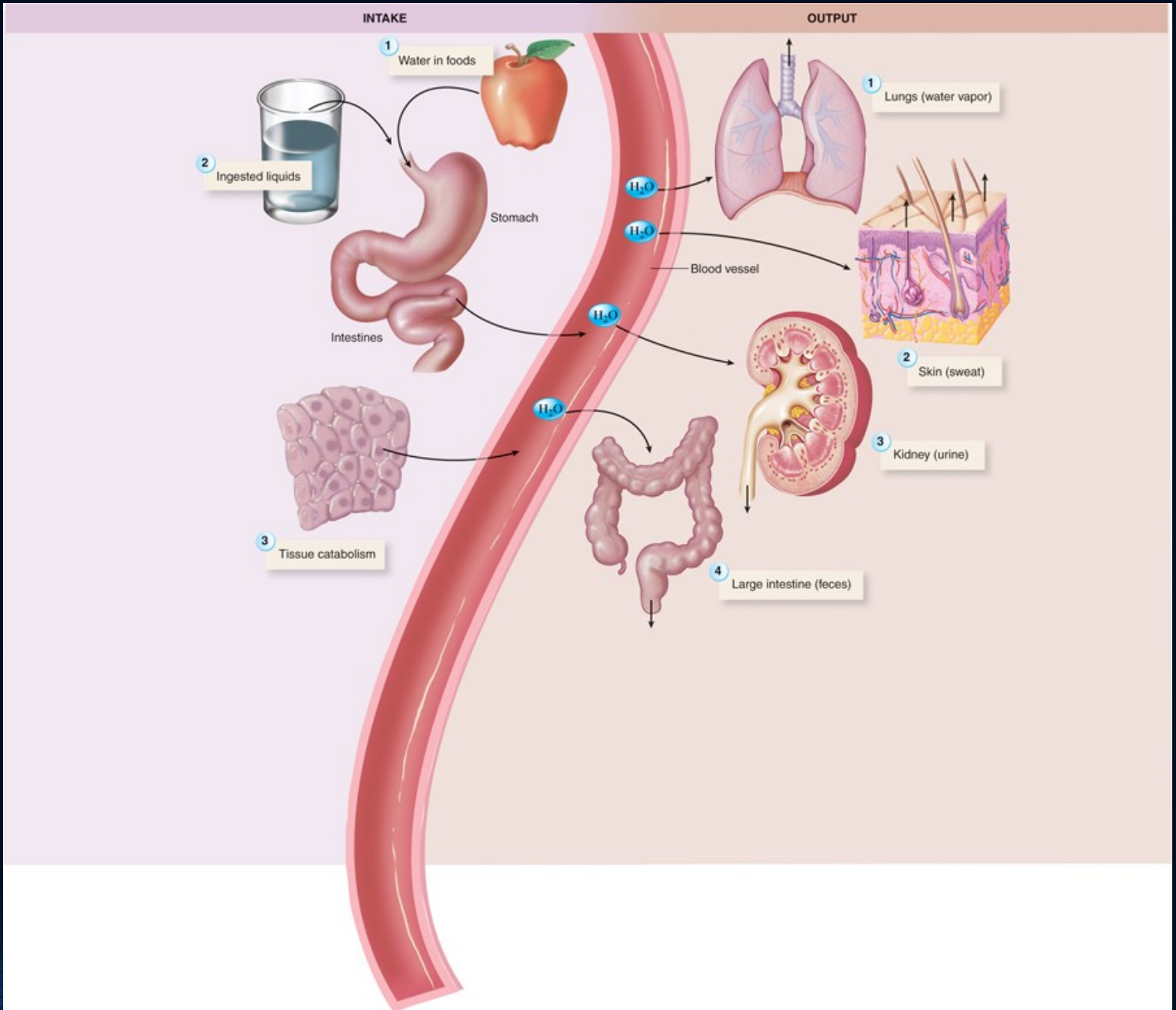
Fluid In / Fluid Out

Fluid IN

- Drinking*
- Eating
- Cellular Metabolism

Fluid OUT

- Urination*
- Sweating
- Ventilation
- Defecation



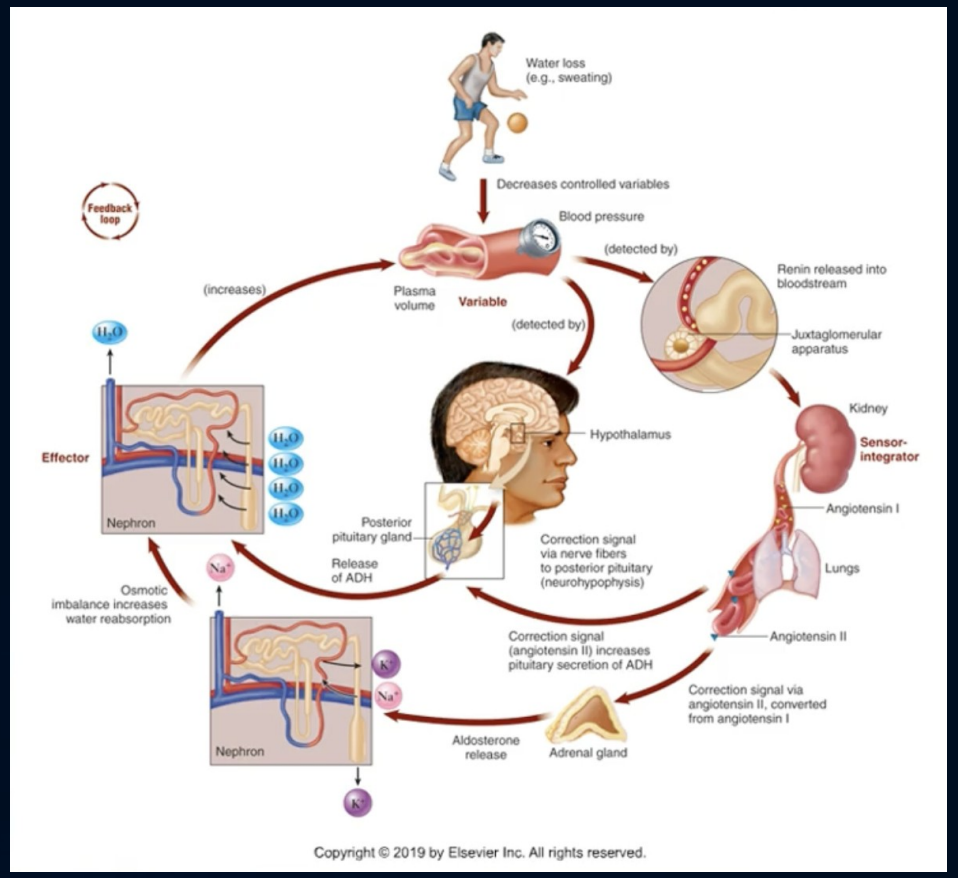
Fluid Homeostasis

- Based on negative feedback mechanisms
- Mechanisms, in addition to affecting fluid volumes, can also affect electrolyte balance and pH balance
- Drinking is the main way in which water can enter the body
- Kidneys do not add water to the body, they conserve the fluid that is already in the body, depending on the needs of the body
- Most of the water released by the body is through excretion by way of the kidneys

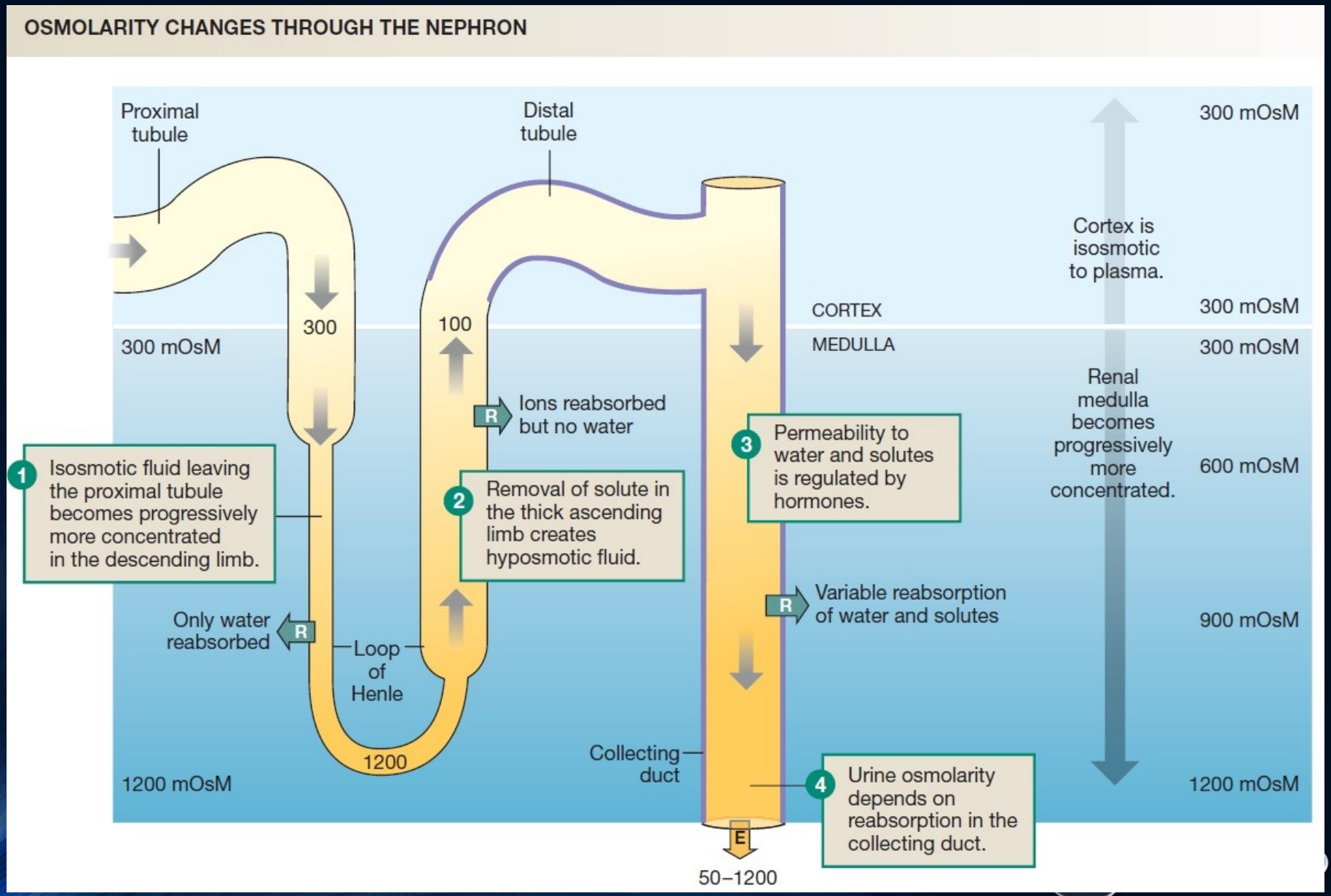
Regulation of ECF Volume and BP

Mechanisms

- Osmotic gradients in the kidney
- Hormonal mechanisms
  - Anti-diuretic hormone
  - Aldosterone
  - Atrial natriuretic peptide
  - Angiotensin II



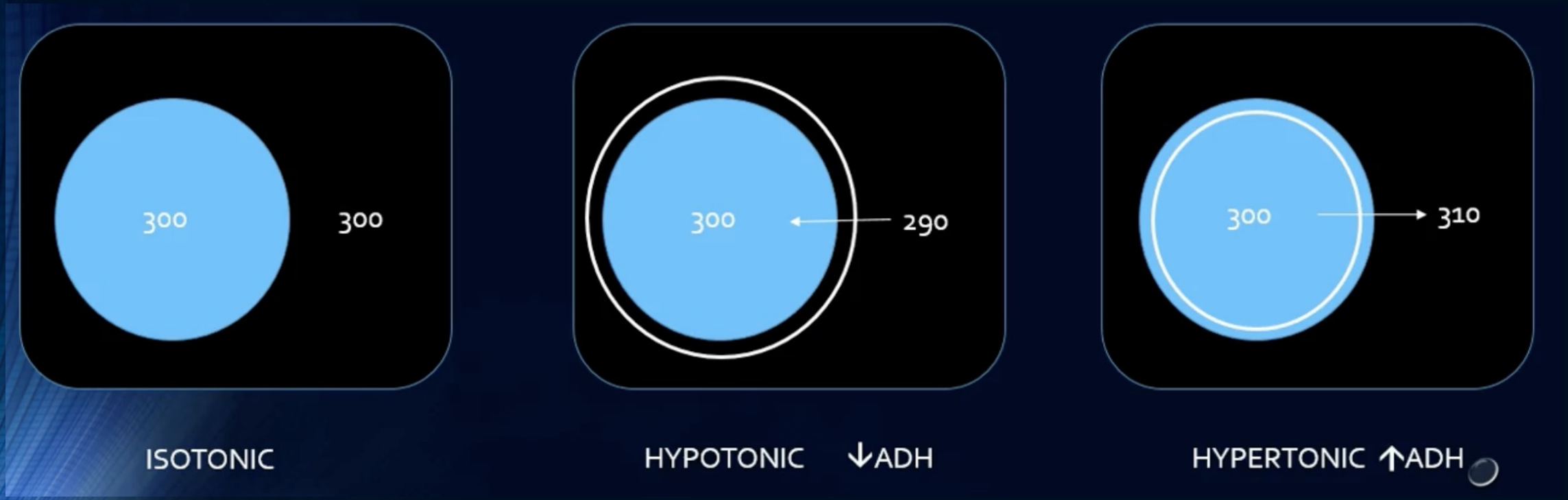
Osmotic Gradients within the Kidney

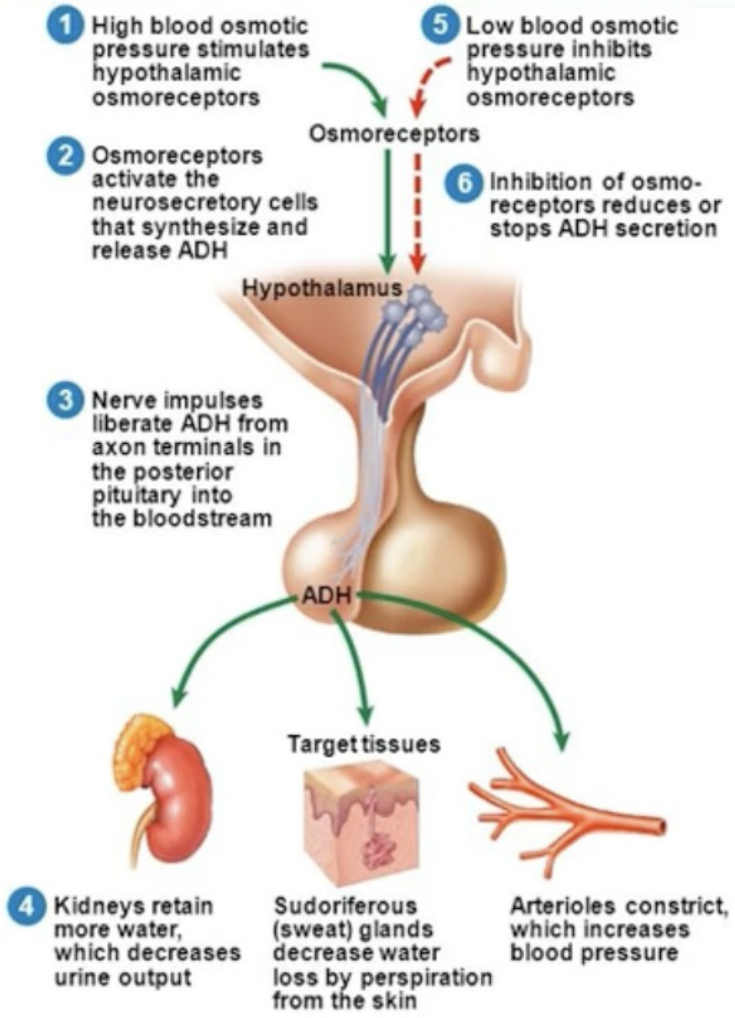


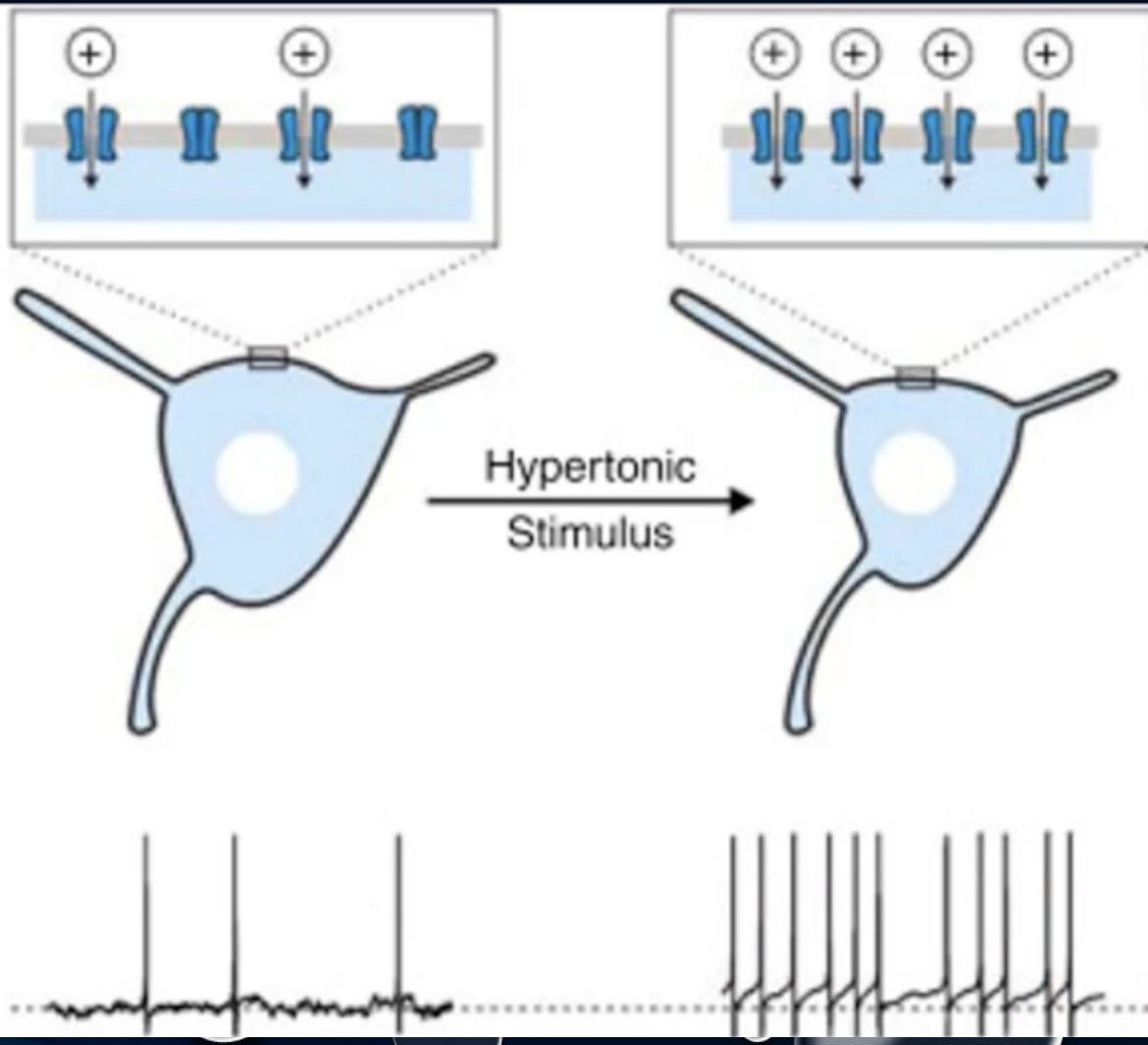
Antidiuretic Hormone / Arginine Vasopressin

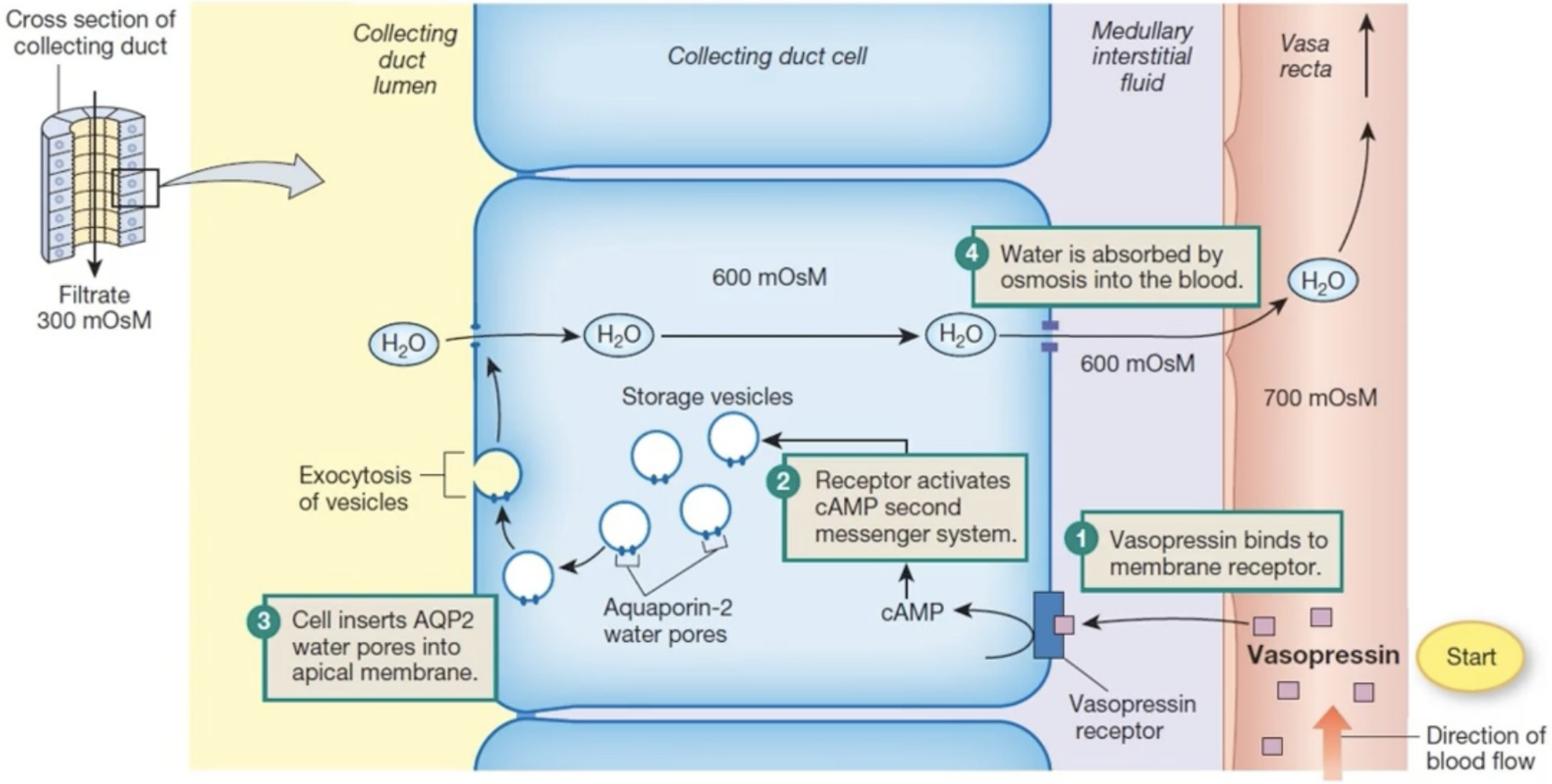
- Stimuli
  - Osmoreceptors
  - Low blood volume / low blood pressure
  - Angiotensin II
  - Atrial stretch receptors and baroreceptors (decrease stretch) → increased ADH secretion
- Actions
  - Increases expression of aquaporin 2 channels on apical side of tubular cells in DCT and CD
  - Without ADH, these channels are not expressed
  - Aquaporin 3 channels on basal surface are always present
- Impact on Fluid Balance and Osmolarity
  - Retains fluid, decreases ECF osmolarity because it only supports reabsorption of water

Isotonic vs Hypotonic vs Hypertonic




1 High blood osmotic pressure stimulates hypothalamic osmoreceptors
5 Low blood osmotic pressure inhibits hypothalamic osmoreceptors
Osmoreceptors
2 Osmoreceptors activate the neurosecretory cells that synthesize and release ADH
6 Inhibition of osmo-receptors reduces or stops ADH secretion
Hypothalamus
3 Nerve impulses liberate ADH from axon terminals in the posterior pituitary into the bloodstream
ADH
Target tissues
4 Kidneys retain more water, which decreases urine output
Sudoriferous (sweat) glands decrease water loss by perspiration from the skin
Arterioles constrict, which increases blood pressure


Hypertonic
Stimulus


Cross section of collecting duct
Filtrate 300 mOsM
Collecting duct lumen
Collecting duct cell
Medullary interstitial fluid
Vasa recta
600 mOsM
4 Water is absorbed by osmosis into the blood.
H_2O
H_2O
H_2O
H_2O
600 mOsM
700 mOsM
Storage vesicles
Exocytosis of vesicles
2 Receptor activates cAMP second messenger system.
1 Vasopressin binds to membrane receptor.
3 Cell inserts AQP2 water pores into apical membrane.
Aquaporin-2 water pores
cAMP
Vasopressin
Start
Vasopressin receptor
Direction of blood flow

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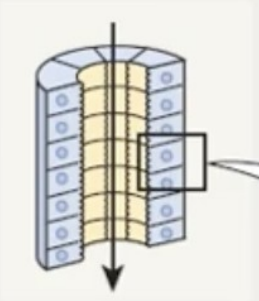
Aldosterone

- Stimuli
  - Hyperkalemia
  - Increased Angiotensin II secretion
- Actions
  - Increases probability that ENaC and ROMK channels will be open on apical side
  - Increases expression of ENaC and ROMK channels, Na+/K+pump
  - Increases rate of catalysis by Na+/K+ pump
- Influence of Fluid Balance and Osmolarity
  - K+ is secreted into lumen; Na+ is reabsorbed
  - Net increase in ion reabsorption causes increase in H2O reabsorption



Lumen of distal nephron

P cell of distal nephron

Interstitial fluid

Blood

2 Hormone-receptor complex initiates transcription in the nucleus.

1 Aldosterone combines with a cytoplasmic receptor.

Transcription

2

mRNA

Aldosterone

3 Translation and protein synthesis makes new protein channels and pumps.

Aldosterone receptor

New channels

New pumps

ATP

4 Aldosterone-induced proteins modulate existing channels and pumps.

K^+ secreted

K^+

K^+

ATP

K^+

Na^+ reabsorbed

Na^+

Na^+

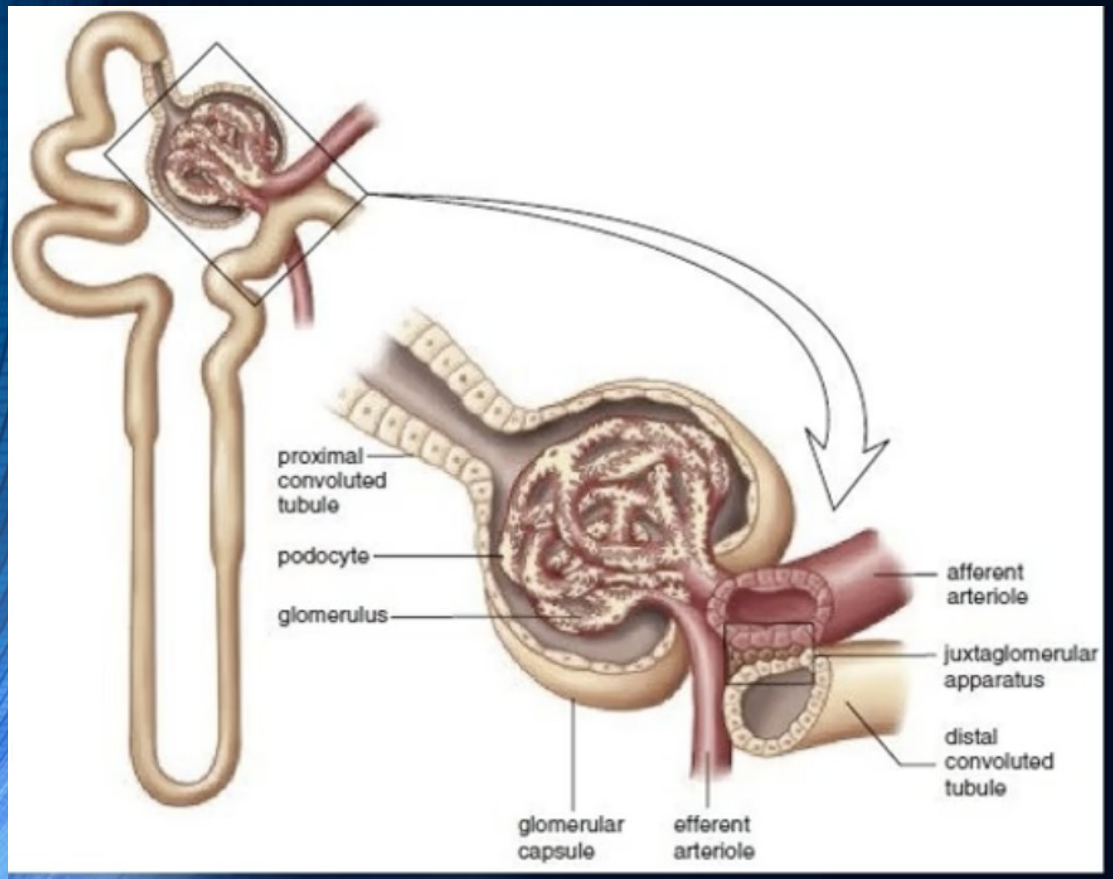
5 Result is increased Na^+ reabsorption and K^+ secretion.

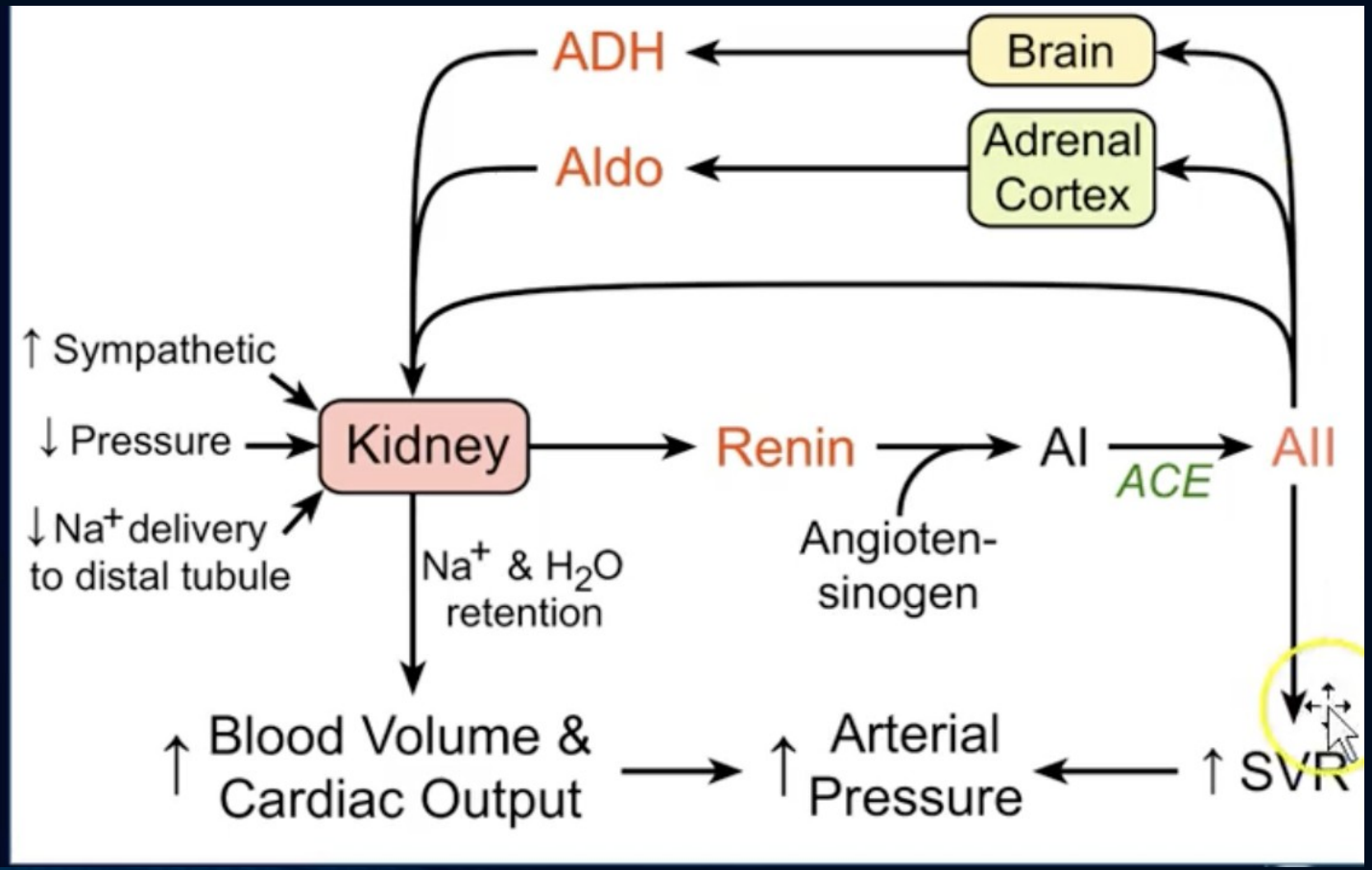
Na^+

Angiotensin II

- Stimuli
  - Reduced blood flow through afferent arteriole / reduced blood pressure
  - Reduced filtrate flow past macula dense
    - Promote release of renin from granular cells
  - Increased sympathetic nervous system activity
- Actions
  - Promotes increased sympathetic activity
  - Vasoconstrictor
  - Promotes secretion of ADH and ALDO
  - Promotes drinking behaviours
- Influence of Fluid Balance and blood pressure

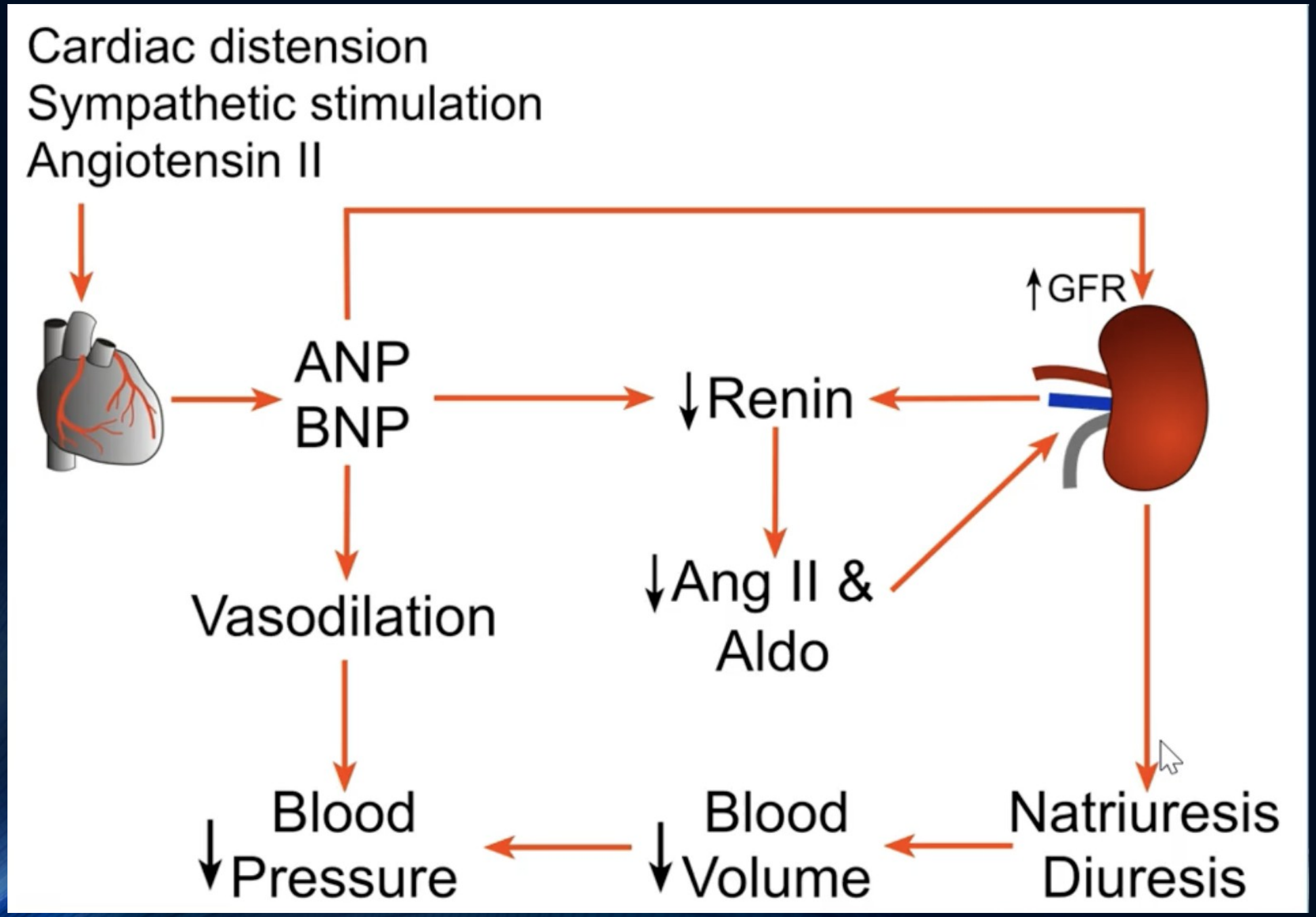
Renin Angiotensin Aldosterone System (RAAS)





Atrial Natriuretic Peptide

- Stimuli
  - Stretch of specialized stretch receptors in atrial myocytes
- Actions
  - Dilates afferent arteriole
  - Suppresses release of renin
  - Decreases probability of ENaC being open in DCT and CD
  - Decreases rate of activity Na+/K+ pump
- Effect on Fluid Balance and Electrolytes
  - Increases GFR, decreases reabsorption of Na+and H2O
  - Decreases production of Ang II – decreases ADH and ALDO release
  - Promotes water loss, decreases fluid volume, decreases BP


Cardiac distension
Sympathetic stimulation
Angiotensin II
ANP
BNP
↑GFR
↓Renin
↓Ang II &
Aldo
Vasodilation
Blood
↓Pressure
Blood
↓Volume
Natriuresis
Diuresis

How does this all fit together?

- Consider a dehydrated person – low blood volume, low BP, high ECF osmolarity
- Osmoreceptors increase secretion of ADH
- Low blood flow past afferent arterioles / low flow of filtrate – renin – Ang II
  - Ang II vasoconstricts, increases ADH and ALDO release
- Low return of blood through atria decreases ANP release
- Low stretch of baroreceptors increases sympathetic stimulation

- Combined – these systems promote reabsorption of water (and Na+) , promote constriction of blood vessels and increased cardiac output